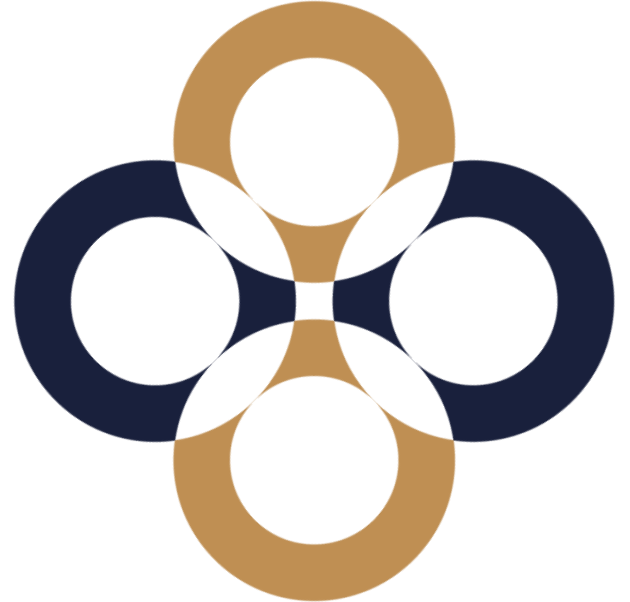


Estimating the cost of capital

Péter Juhász, PhD, CFA



Matching the CF and the cost of capital

Currency

**Effects of
financing
(TS)**

Inflation

Maturity

Risk

Return of similar investments

How is that similar?

How do we know it is similar?

Measuring risk, calculating required rates

CAPM

Alternative methods (APT)



Whose risk is to be measured?

Investor	Risk	Metric	Requirement	Firm value
Private investor (one firm)	Total	Std deviation	40%	$100/0.40= 250$
Venture capital (one sector)	Sector portfolio +	Sector index beta	25%	$100/0.25= 400$
Hungarian investment found	Country portfolio +	Country index beta	15%	$100/0.15= 666$
Investment bank	Global portfolio +	Global index beta	10%	$100/0.10=1000$

Measuring the cost of capital

Cost of Debt

Historical risk premiums

Credit rating premiums

CAPM and debt beta

Cost of Equity

CAPM

APT

Implied return

Build-up method

Cost of capital inputs 1 – Risk free rate

Free only from repayment risk; other (e.g. market) risks still exist.

Real or
nominal?

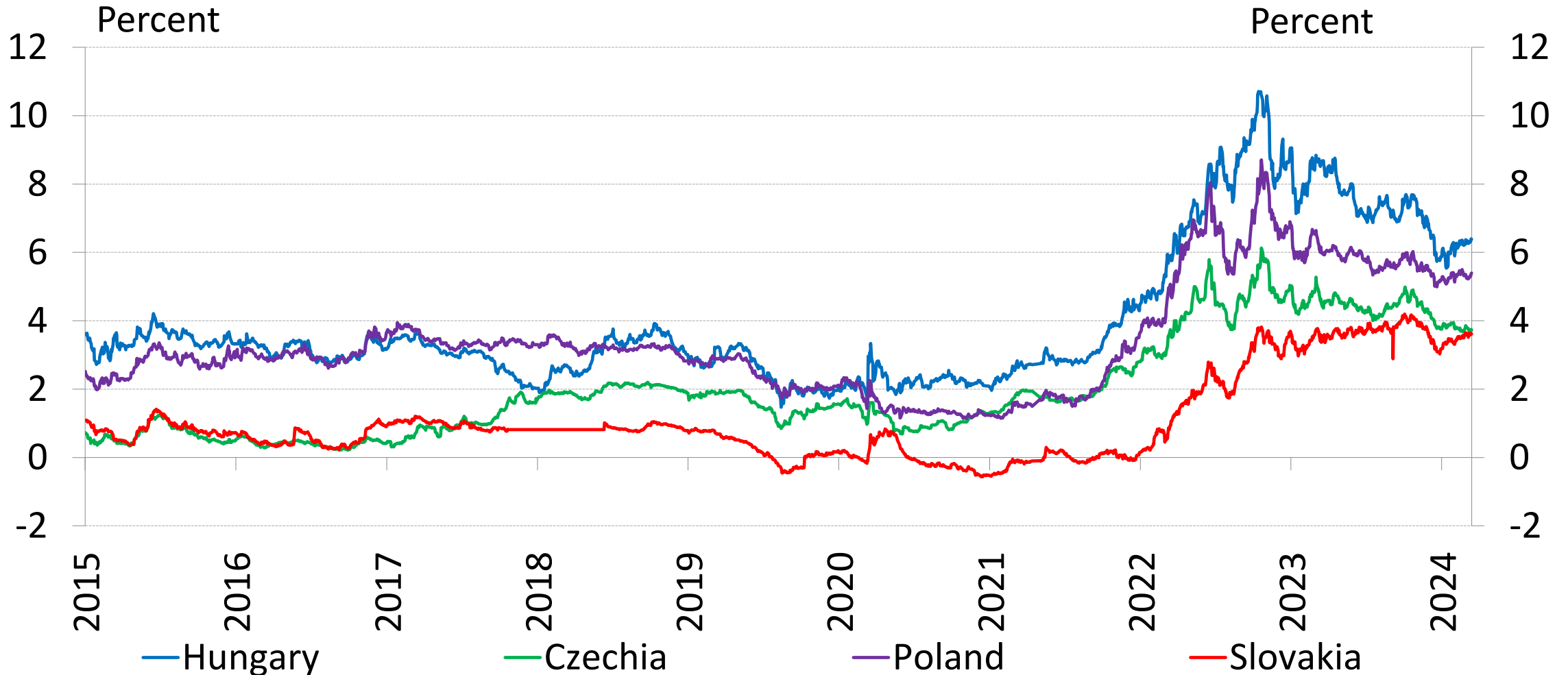
Which country?
(Switzerland vs
Greece)

Yield curves
(spot/forward,
never constant)

Exchange rate
predictions
(parities)

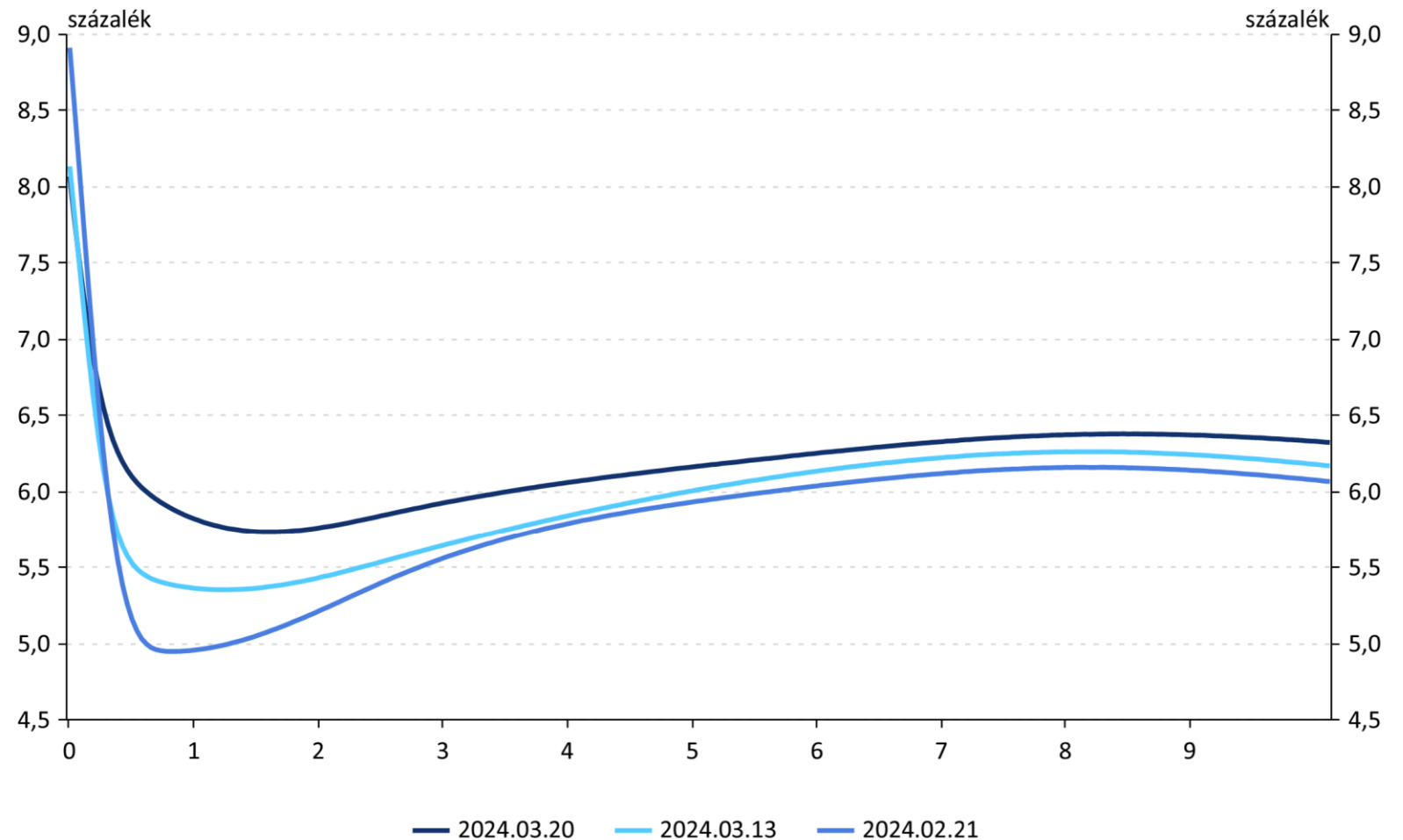
Consistent with
inflation
predictions?

10-year government yields in CEE countries



Estimating the forward rates

3-months interbank rates



Source:
MNB, 26 March 2024

Cost of capital inputs 2 – Cost of debt

Former and
existing debt
contracts

Implied debt return
of own or similar
firms' bonds

Historical
regressions
(β_D)

Credit rating
(risk free rate +
premium)

S&P credit rating premiums*

Rating	2010	2012	2014	2016	2018	2020	2022	2024
AAA	0.50%	0.65%	0.40%	0.75%	0.54%	0.63%	0.69%	0.59%
AA	0.75%	1.15%	0.70%	1.00%	0.72%	0.78%	0.85%	0.70%
A+	1.00%	1.30%	0.85%	1.10%	0.90%	0.98%	1.07%	0.92%
A	1.25%	1.40%	1.00%	1.25%	0.99%	1.08%	1.18%	1.07%
A-	1.50%	1.65%	1.30%	1.75%	1.13%	1.22%	1.33%	1.21%
BBB	2.00%	2.50%	2.00%	2.25%	1.27%	1.56%	1.71%	1.47%
BB+	3.50%	3.75%	3.00%	3.25%	1.98%	2.00%	2.31%	1.74%
BB	4.00%	4.75%	4.00%	4.25%	2.38%	2.40%	2.77%	2.21%
B+	4.25%	5.50%	5.50%	5.50%	2.98%	3.51%	4.05%	3.14%
B	5.25%	6.00%	6.50%	6.50%	3.57%	4.21%	4.86%	3.61%
B-	5.50%	6.75%	7.25%	7.50%	4.37%	5.15%	5.94%	5.24%
CCC	8.50%	8.75%	8.75%	9.00%	8.64%	8.20%	9.46%	8.51%
CC	10.00%	9.50%	9.50%	12.00%	10.63%	8.64%	9.97%	11.78%
C	12.00%	10.50%	10.50%	16.00%	13.95%	11.34%	13.09%	17.00%
D	15.00%	12.00%	12.00%	20.00%	18.60%	15.12%	17.44%	20.00%

*Huge manufacturing firms, USD

Source:
Damodaran,
January 2010-2024

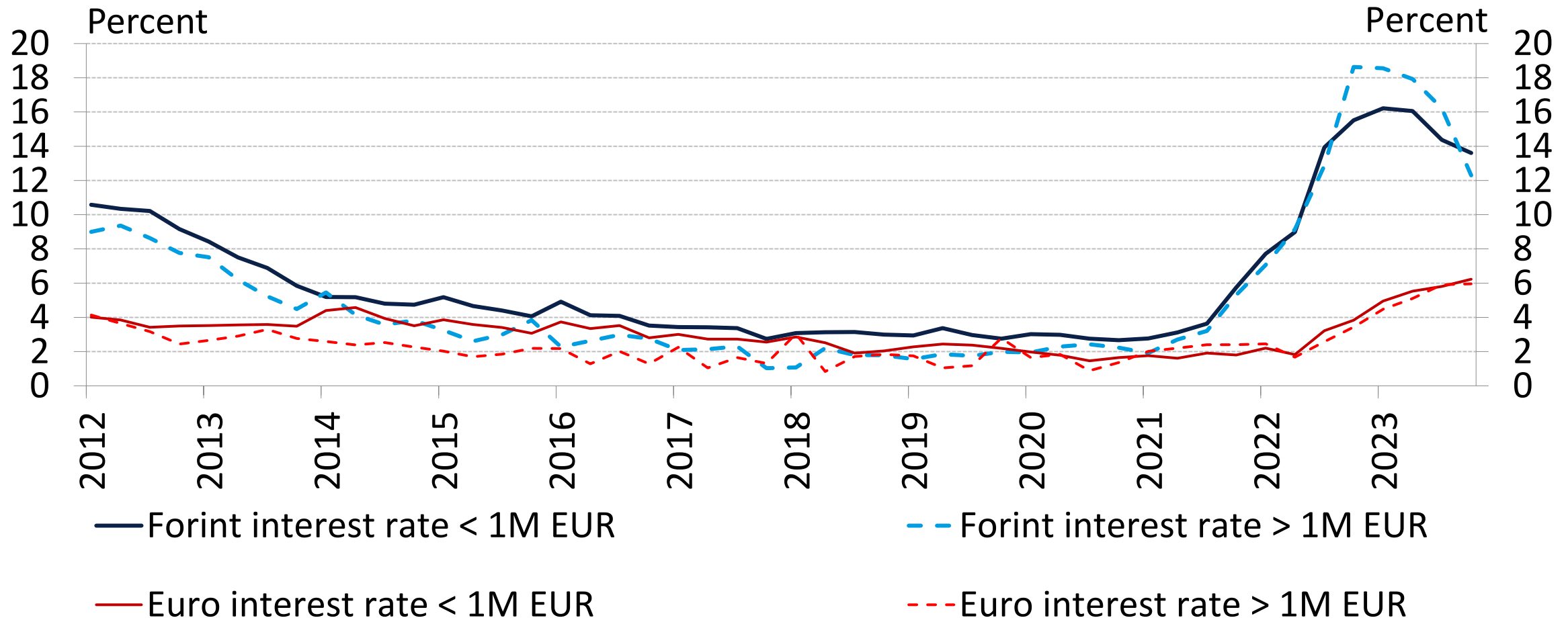
Rating	1 y	2 y	3 y	5 y	7 y	10 y	30 y
Aaa/AAA	14	16	27	40	56	68	90
Aa1/AA+	22	30	31	48	64	77	99
Aa2/AA	24	37	39	54	67	80	103
Aa3/AA-	25	39	40	58	71	81	109
A1/A+	43	48	52	65	79	93	117
A2/A	46	51	54	67	81	95	121
A3/A-	50	54	57	72	84	98	124
Baa1/BBB+	62	72	80	92	121	141	170
Baa2/BBB	65	80	88	97	128	151	177
Baa3/BBB-	72	85	90	102	134	159	183
Ba1/BB+	185	195	205	215	235	255	275
Ba2/BB	195	205	215	225	245	265	285
Ba3/BB-	205	215	225	235	255	275	295
B1/B+	265	275	285	315	355	395	445
B2/B	275	285	295	325	365	405	455
B3/B-	285	295	305	335	375	415	465
Caa/CCC+	450	460	470	495	505	515	545
USA gov. bond rate	4.74	4.71	4.68	4.63	4.60	4.59	4.56

Reuters credit rating premiums*

*Banks. USD.
January 2006

Source:
Bondsonline.com

Hungarian corporate debt rates*



*New lendings

Source: Inflation report, MNB, December 2023

Cost of capital inputs 3 – Tax rate

Nominal (if no information available)

Effective (historic)

Marginal (extensional investments)

Combined (growing firm)

We need predictions.

Corporate tax rate in the CEE region

%	2008	2010	2012	2014	2016	2018	2020	2021	2022	2023
Albania	10	10	10	15	15	15	15	15	15	15
Bosnia and Herzegovina	10	10	10	10	10	10	10	10	10	10
Bulgaria	10	10	10	10	10	10	10	10	10	10
Czech Republic	21	19	19	19	19	19	19	19	19	19
Estonia	21	21	21	21	20	20	20	20	20	20
Belarus	24	24	18	18	18	18	18	18	18	18
Croatia	20	20	20	20	20	18	18	18	18	18
Poland	19	19	19	19	19	19	19	19	19	19
Hungary	16	19	19	19	19	9	9	9	9	9
Russia	24	20	20	20	20	20	20	20	20	20
Romania	16	16	16	16	16	16	16	16	16	16
Serbia	10	10	10	15	15	15	15	15	15	15
Slovakia	19	19	19	22	22	21	21	21	21	21
Slovenia	22	20	18	17	17	19	19	19	19	19
Ukraine	25	25	21	18	18	18	18	18	18	18

Points of reference

	2010	2012	2014	2016	2018	2020	2022	2023
Austria	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	24,00%
Cyprus	10.00%	10.00%	12.50%	12.50%	12.50%	12.50%	12.50%	12,50%
United States	40.00%	40.00%	40.00%	40.00%	27.00%	27.00%	25.00%	25,00%
United Kingdom	28.00%	24.00%	21.00%	20.00%	19.00%	19.00%	19.00%	25,00%
France	33.33%	33.33%	33.33%	33.30%	33.00%	33.00%	25.00%	25,00%
Greece	24.00%	20.00%	26.00%	29.00%	29.00%	29.00%	22.00%	22,00%
Ireland	12.50%	12.50%	12.50%	12.50%	12.50%	12.50%	12.50%	12,50%
Japan	40.69%	38.01%	35.64%	30.86%	30.86%	30.86%	23.20%	23,20%
China	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	25.00%	25,00%
Germany	29.41%	29.48%	29.58%	29.72%	30.00%	30.00%	30.00%	30,00%
North America aver.	35.50%	33.00%	33.25%	33.25%	26.75%	26.75%	26.75%*	
Asia average	23.96%	22.89%	21.91%	21.92%	21.21%	21.21%	21.09%*	
EU average	23.04%	22.60%	21.34%	22.09%	21.29%	21.29%	21.12%*	
Global average	24.69%	24.40%	23.57%	23.62%	24.05%	24.03%	23.79%*	

Source: KPMG. quoted by: Damodaran, 2024 *Avg for 2021

	2008	2010	2012	2014	2016	2018	2020	2022	2023
Bahrain	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
United Arab Emirates	55.00%	55.00%	55.00%	55.00%	55.00%	55.00%	55.00%	0.00%	0.00%
Kuwait	55.00%	15.00%	15.00%	15.00%	15.00%	15.00%	15.00%	15.00%	15.00%

Source: KPMG, quoted by: Damodaran, 2024



Cost of capital inputs 4 – Cost of equity

Interview (What do owners want?)

Expert opinion (required rates for other firms)

Regression methods: CAPM, APT

Built-up method

Cost of equity contains two types of risk:

Risk of operation:

$$r_A = r_f + \beta_A(r_M - r_f)$$

Risk of financing:

$$\beta_E = \beta_A + \frac{D}{E}(\beta_A - \beta_D)$$

$$r_E = r_A + \frac{D}{E}(r_A - r_D)$$



$$r_E = r_f + \beta_E(r_M - r_f)$$

Historical regressions:

$$r_{\text{Asset}} = a + b * r_M$$
$$b \text{ (sloep)} = \text{covar}(r_{\text{Asset}}, r_M) / \sigma_M^2 = \text{„}\beta_E\text{”}$$

Market
portfolio?

Time frame?

Yield type?

Currency?

Exchange
rate
corrections?

Steps of beta estimation

Collect industry player data (r_E , β_E , r_D , β_D)

Estimate unleveraged cost of capital individually

Calculate the average of individual values (r_A , β_A)

Re-leverage r_A and β_A in line with the target's financing structure

Market risk premium applied*

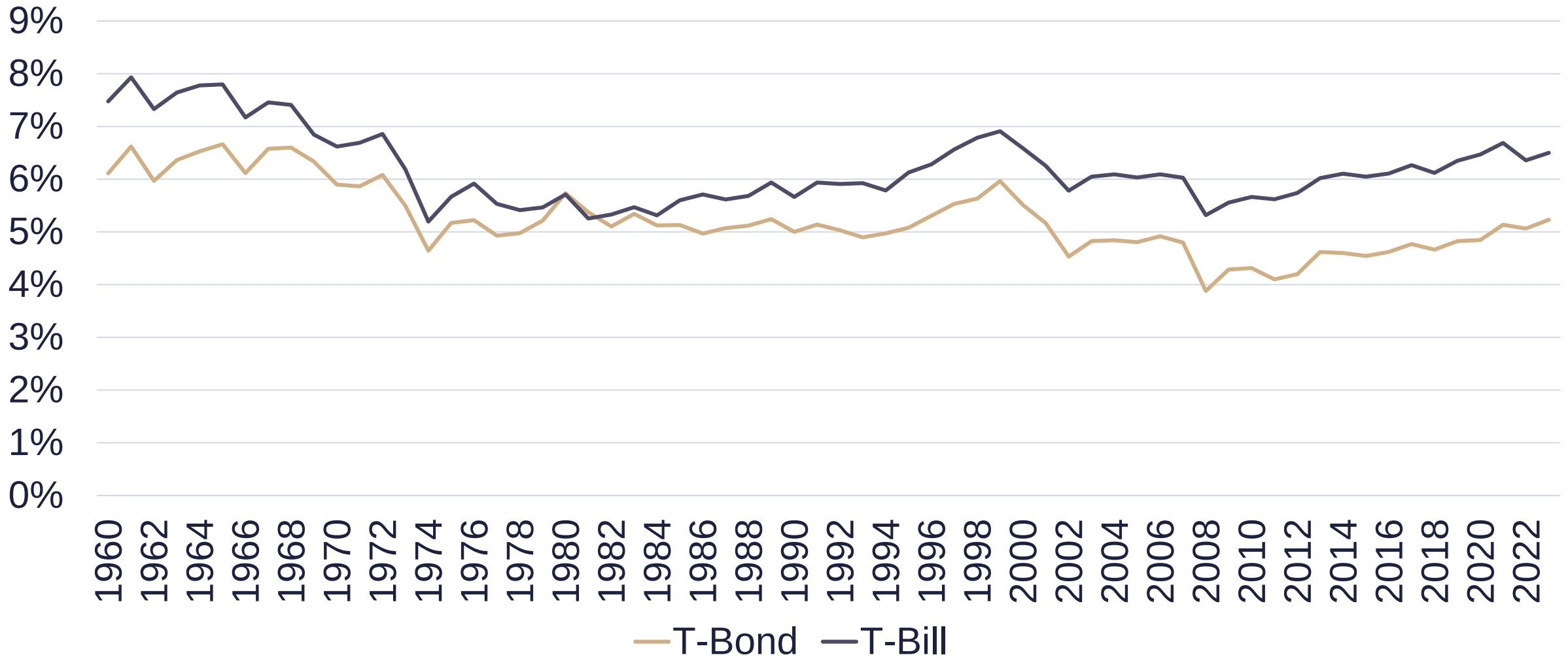
Country	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Austria	6.0	5.8	5.5	5.6	5.3	6.6	6.4	6.3	5.9	6.0	6.6	6.6	5.9
Belgium	6.0	6.0	5.5	5.4	5.5	6.6	6.4	6.3	6.0	6.0	7.0	7.0	5.5
UK	5.0	5.0	5.0	5.0	5.0	6.2	5.9	6.0	5.8	5.7	6.0	6.0	5.6
France	6.0	5.9	6.0	5.5	5.5	6.7	6.4	6.1	6.0	5.7	6.3	6.3	5.6
Greece	6.0	16.5	7.4	15.0	12.4	17.6	16.9	15.3	14.6	6.4	12.2	12.2	6.0
Germany	5.0	5.0	5.4	5.1	5.0	5.9	5.2	5.7	5.7	5.5	5.9	5.9	5.6
Italy	5.5	5.5	5.5	5.2	5.5	6.7	6.4	6.1	6.1	6.0	6.0	6.0	6.0
Portugal	5.9	8.5	6.5	5.5	8.0	8.0	7.7	7.7	6.9	6.5	6.8	6.8	6.0
Spain	5.5	6.0	5.9	5.5	6.0	6.8	6.2	6.4	6.4	6.4	6.3	6.3	6.0
USA	5.5	5.0	5.5	5.3	5.0	5.7	5.2	5.5	5.4	5.5	5.5	5.5	5.5
Switzerland	5.5	5.0	5.7	5.0	5.0	7.5	7.2	6.0	5.9	5.1	5.9	5.9	5.3
Hungary	7.0	8.7	8.9	8.9	8.0	8.6	8.4	8.5	7.1	7.1	9.0	9.0	6.0

* $r_M - r_f$, median

Source: Fernandez, 2012-2024

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Historical market premium*



Source: Damodaran, March 2024

Historical market premium*

* $r_M - r_f$, %, USD, difference between **S&P500** and US government bonds, between the given year and **2013**, geometric average of yearly rates

Gov. bond type	2002	2004	2006	2008	2009	2010	2011	2012
Long term	1.70	2.50	2.30	0.90	11.40	7.60	6.00	13.60
Mid term	3.10	3.80	3.70	2.70	13.40	9.70	8.00	15.40
Short term	4.40	4.90	4.70	4.10	14.80	11.00	9.00	15.90

Source: Ibbotson SBBI, 2013

Fundamental beta

Industry (cycles, basic vs luxury) – „ β_{industry} ”

Operative risks (cost structure: variable/fixed cost,

$\Delta \text{EBIT} / \Delta \text{Sales} - \text{DOL}$) – „ β_{strategy} ”

– $\Phi(\beta_{\text{industry}}, \beta_{\text{strategy}}) \Rightarrow \text{„}\beta_A\text{”}$

Financing risks (taxation, D/E) $\Rightarrow \text{„}\beta_E\text{”}$

Cost of capital inputs 6 – Leverage ratio

Historical
data

Predicted
trend

Target
(announced)
level

Iteration

Industry betas (Western EU)

Industry	N	β_E	β_A	D/V
Semiconductor	40	1.40	1.24	12.23%
Apparel	110	1.00	0.90	12.42%
Air Transport	34	1.34	0.76	47.55%
Steel	59	1.04	0.75	31.08%
Oil/Gas (Production and Exploration)	91	0.84	0.53	39.98%
Green & Renewable Energy	66	0.83	0.50	42.41%
Farming/Agriculture	50	0.80	0.45	47.05%
Utility (Water)	12	0.76	0.38	52.43%
Food Wholesalers	14	0.76	0.32	62.49%

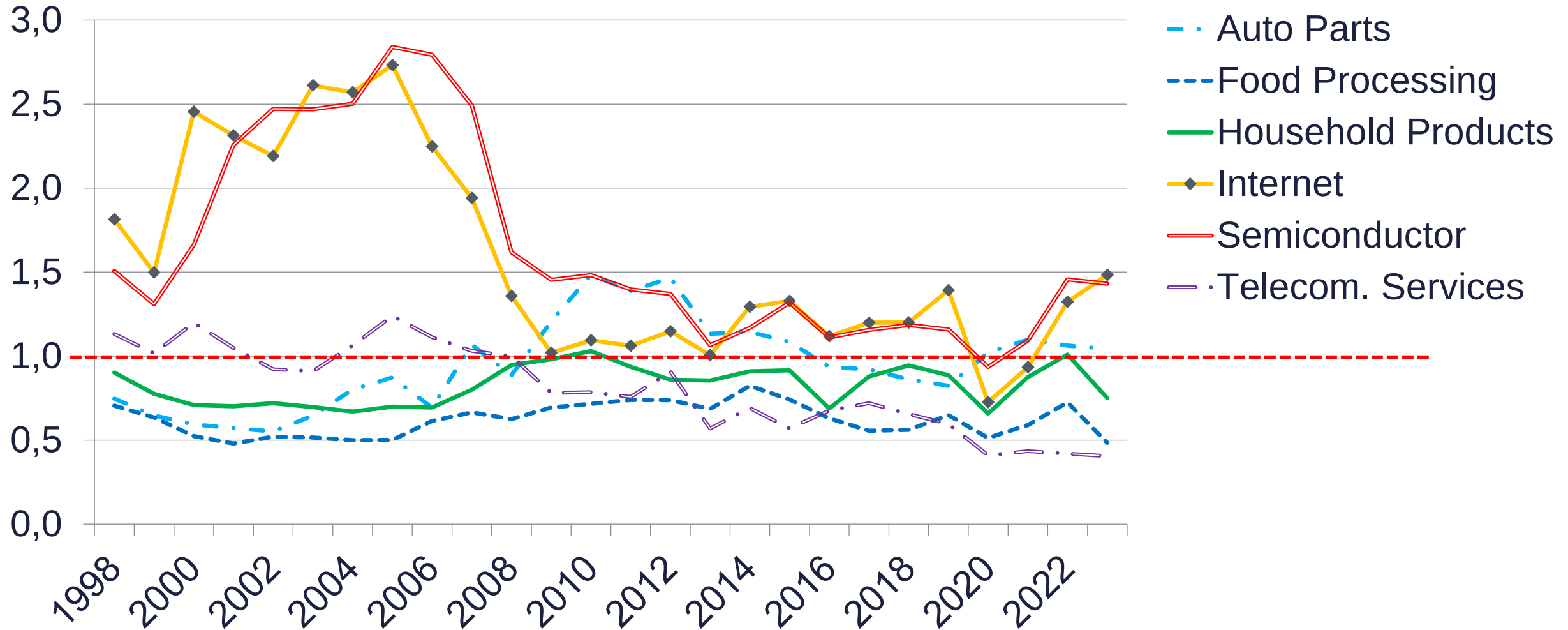
Stability of β_A^* over time



*in the USA

Source: Damodaran online, March 2023

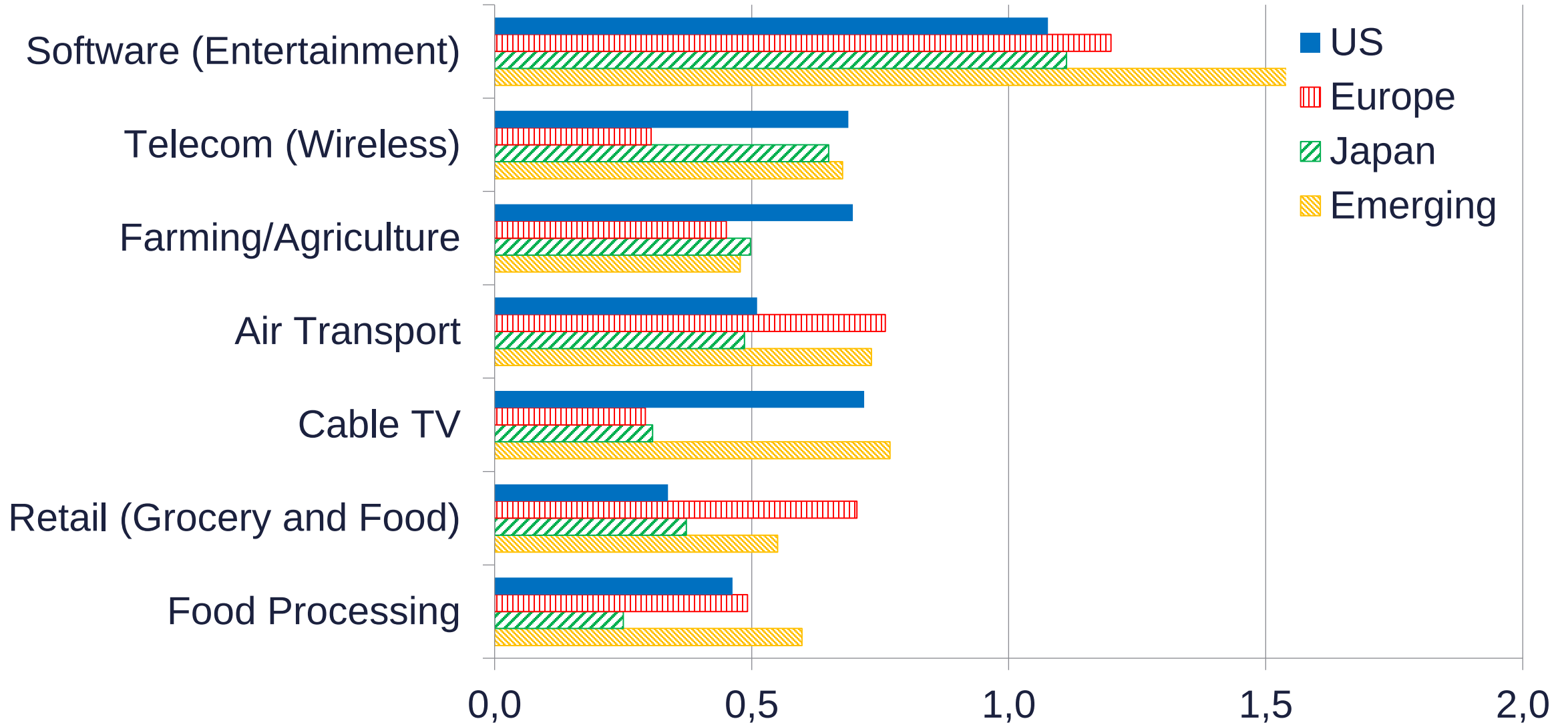
Stability of β_A^* over time



*in the USA

Source: Damodaran online, March 2024

Geographic stability of β_A



Estimating the cost of equity using the build-up method

Estimating required rate on equity:

$$\varepsilon(r_A) = r_f + \text{MRP} + \text{RP}_I + \text{RP}_S + \text{RP}_O$$

where

$E(r_E)$ = required rate of equity,

r_f = risk-free rate of return,

MRP = market risk premium,

RP_I = industry risk premium,

RP_S = size risk premium,

RP_O = other firm specific risk premium.

Government bond rates of developed countries	1.0%
Country risk premium of Hungary	1.5%
Inflation difference	2.0%
Risk free rate (r_f)	4.5%
Market risk premium (MRP: $r_M - r_f$)	6.5%
Industry premium	1.5%
Size and other firm specific premiums	2.0%
Required return of assets (r_A)	14.5%

No need for betas.

Stability of size premium*

*%, USD, the yield premium of the given sized firms compared to huge companies, between the given year and **2013**.

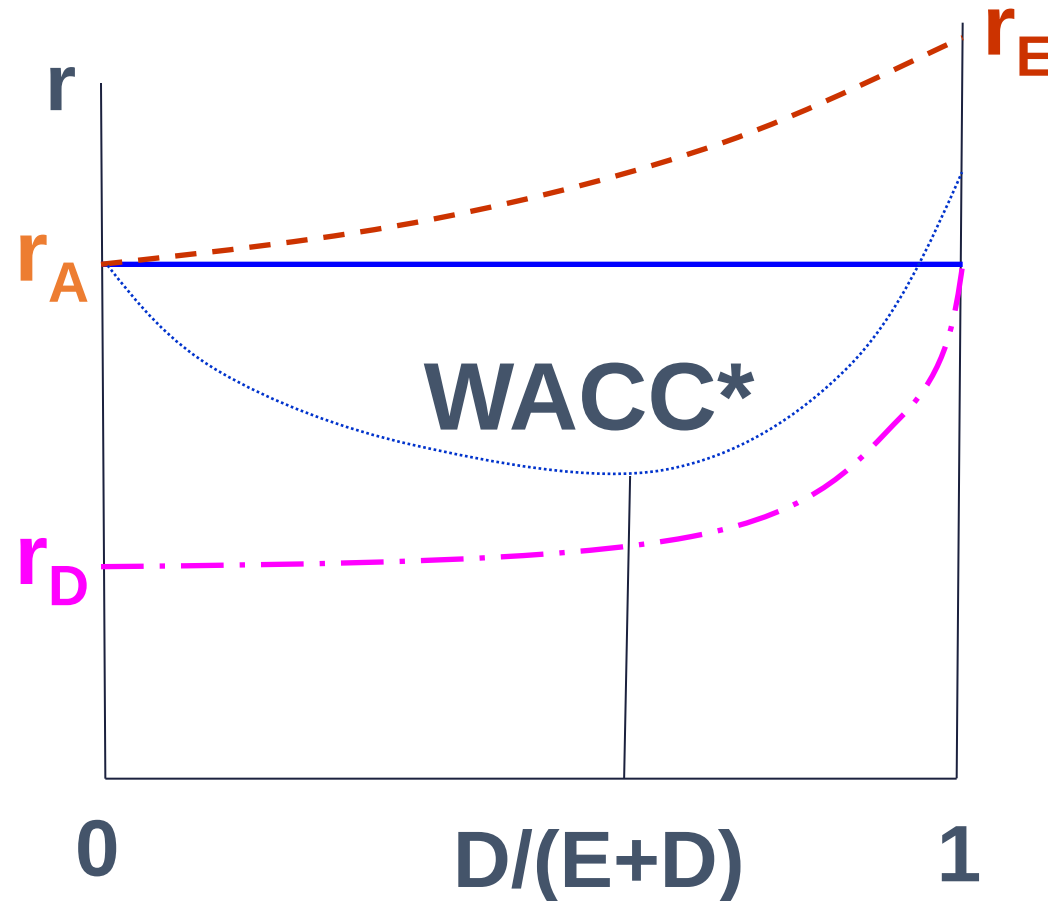
Capitalisation**	2001	2002	2003	2004	2005	2006	2007	2008
Medium	5.0	4.5	4.0	3.5	3.1	2.7	3.7	4.6
Small	7.0	4.9	4.4	3.2	2.5	2.7	3.6	5.5
Micro	10.8	7.1	5.9	2.1	1.9	2.4	3.0	6.4

** Categories include NYSE, NYSE Amex, and NASDAQ firms' 3-5, 6-8, 9-10 deciles.

Source: Ibbotson SBBI, 2013.

$$WACC = \frac{D}{V} * r_D * (1 - t) + \frac{E}{V} * r_E$$

But this formula misses the effects of financial distress.



* With financial distress effects

Can you reach the optimum?

Curves are not continuous,

Only your direct neighbourhood can be tracked.

The amount of debt could be limited.

Curves may deviate across financiers.

Curves change over time.



**Thank you
for your attention.
Any questions?**

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